



Title of Lecture: **Physics of diagnostic Radiology**

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References:

- ❑ 2014- Diagnostic Radiology Physics.
- ❑ 2023_Khan's the Physics of Radiation Therapy.
- ❑ 2023_ Introduction of physics in Modern Medicine.
- ❑ 2019-Diagnostic Ultrasound Physics and Equipment.
- ❑ 2020- Magnetic Resonance Imaging (MRI).
- ❑ 2020_@ Radiological Physics with Cases (Important).

Chapter 1.

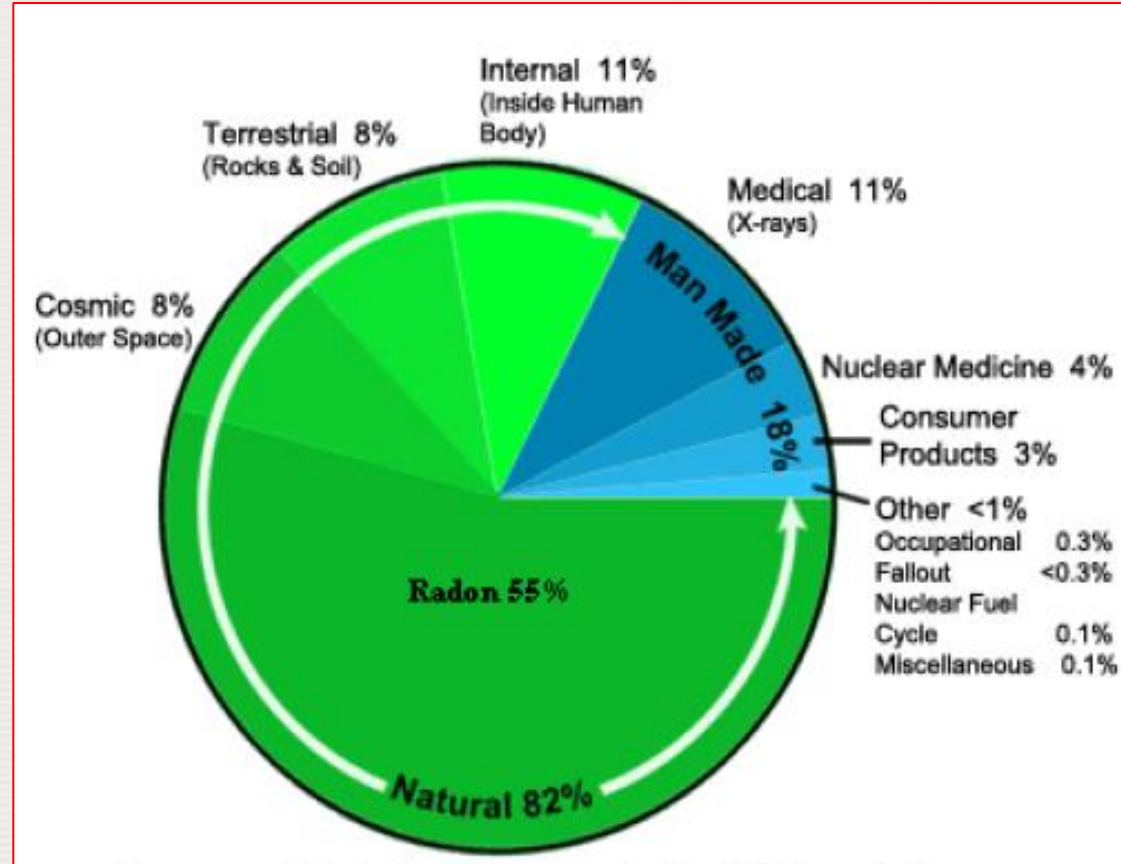
Objectives: To familiarize the student with the principles of the construction and operation on
Physics of Diagnostic imaging systems

Chapter 1. TABLE OF CONTENTS

- 1.1 Source of Radiation.
- 1.2 Measurements of radiation dose.
- 1.3 The principles of the operation of X-ray imaging systems.
- 1.4 The principles of the operation of fluoroscopic imaging systems.
- 1.5 The requirements and principles of imaging of the breast using (X- ray mammography).
- 1.6 Dental radiography, Mobile Radiography and tomosynthesis
- 1.7 DUAL-ENERGY X-RAY ABSORPTIOMETRY**
- 1.8 CT scanning principles, acquisition and reconstruction, technology and image quality .
- 1.9 MRI (Magnetic Resonance Imaging)

1.1 Sources of Radiation

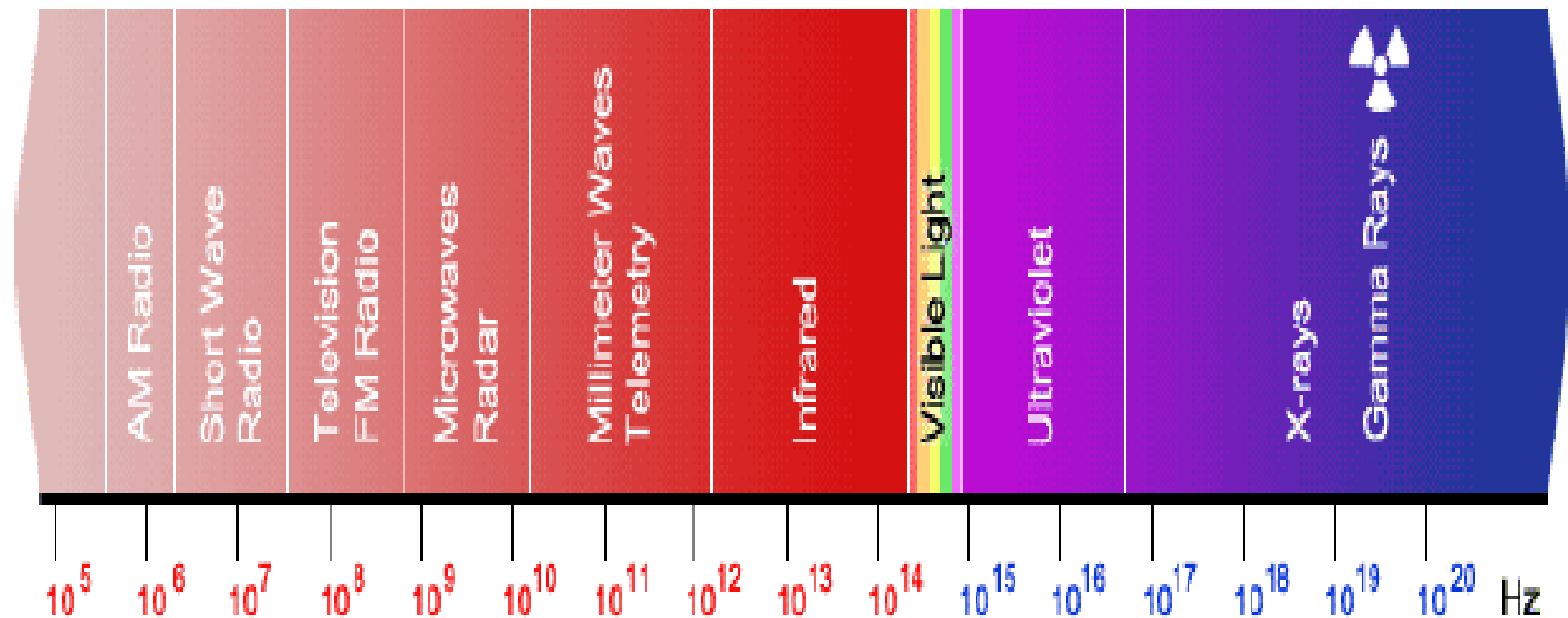
- ☐ Natural (82%)
- ☐ Man made: (18%)
 - X-rays
 - CT scans
- ☐ Nuclear medicine
- ☐ radiation oncology



When you are exposed to radiation, your body absorbs a measurable dose

Electromagnetic Radiation

Types of Electromagnetic Radiation



Non-Ionizing Radiation

Ionizing Radiation

1.2 Measurement of Radiation Dose

Radiation Measurements

	Exposure	Absorbed Dose	Dose Equivalent	Radioactivity
Common Units	roentgen (R)	rad	rem	curie (Ci)
SI Units	coulomb/kilogram (C/kg)	gray (Gy)	sievert (Sv)	becquerel (Bq)

**SI – International System of Units

The Roentgen : is used to express the amount of **gamma radiation exposure**

RAD: Relates different types of radiation (**alpha, beta, gamma and neutron**) to the energy they impart

REM: Relates the dose of any radiation to the **biological effect** of that dose



- ☐ What needs to be known for **medical treatment**
- ☐ Intensity of **exposure**
- ☐ **Time or duration** of exposure

1.3 X-RAYS

- ☐ **X-rays** are high energy,
- ☐ high frequency
- ☐ **electromagnetic** waves
- ☐ They are generated in Xray tubes
- ☐ They are used in medicine for **Diagnostic**
- ☐ and **Therapeutic**
- ☐ applications



- These three basic phenomena form the bases of three important X-ray methods: **the absorption** technique, which is the basis of radiographic analysis; **the scattering** effect, which is the basis of X-ray diffraction; and the **fluorescence** effect, which is the basis of



Discovering the Inside Story:

INNOVATIONS IN MEDICAL IMAGING AND RADIATION THERAPY

This National Radiologic Technology Week, travel back in time and relive the history of the radiologic technology profession and the American Society of Radiologic Technologists through ASRT Museum and Archives exhibits.



Wilhelm Conrad Roentgen

1895
German physicist Wilhelm Conrad Roentgen discovers the x-ray and within a year it is hailed as a medical miracle.

1896
First clinical use of x-rays by physicians.

1903
First use of radiation to treat cancer.

1904
Thomas Edison's assistant C.M. Dally dies of radiation exposure; called "a martyr to science."

1913
Known for his many contributions to x-ray technology, William Coolidge invents the heated cathode x-ray tube, enabling external beam radiotherapy.

1914
Marie Curie designs vehicles to carry x-ray equipment to the battlefield in World War I.

1920
American Association of Radiological Technicians established; known today as the American Society of Radiologic Technologists.

1927
Thirty-seven percent of all radiologic technologists are nurses.

1941
The U.S. military trains thousands of servicemen as radiographers to provide care for wounded soldiers overseas during World War II.

1950s
Radiographers take chest x-rays using mobile x-ray units to screen for tuberculosis.

1964
Radiation therapy becomes a specialized field.

1971
First CT scan.

1983
ASRT opens new national office in Albuquerque, New Mexico.

1977
Drs. Raymond V. Damadian, Lawrence Minkoff and Michael Goldsmith's MRI scanner makes the first-ever MR image.

1979
Allan Cormack and Godfrey Hounsfield awarded Nobel Prize in Medicine for invention of computed tomography.

2015
ASRT Museum and Archives opens.

2011
First clinical 3-D mammography exam performed in the U.S.



Photo: JCNAB

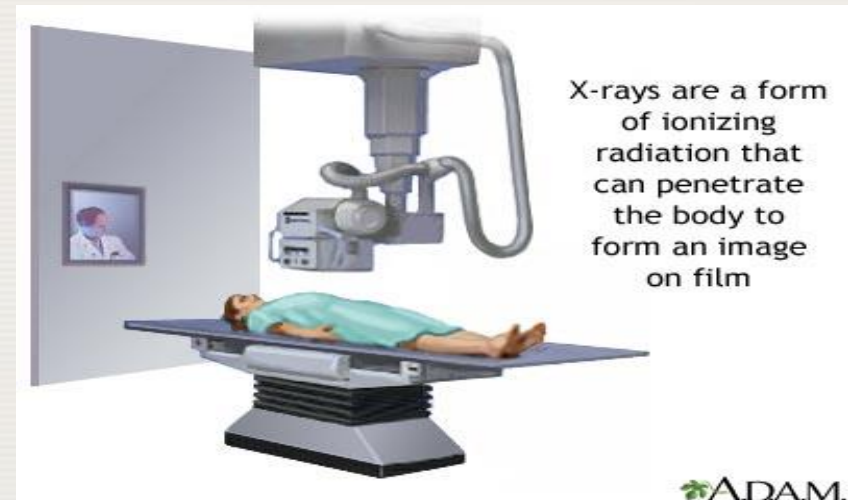
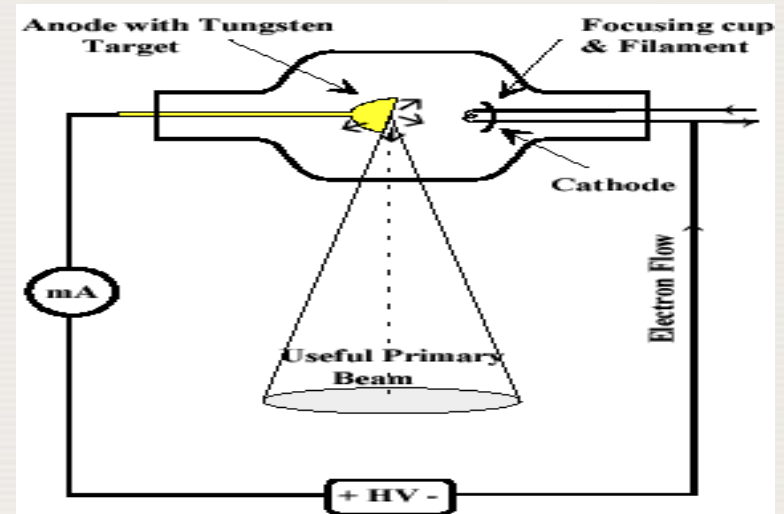
Photos: 1913, 1927, 1941, 2015: Kip Malone

asrt Museum and Archives

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1.3 X-ray Tube

- ❑ A large current passes through the **cathode** which creates a heating effect
- ❑ This causes electrons to be emitted by **thermionic emission**
- ❑ The electrons are accelerated towards the **tungsten target** by a very large potential difference
- ❑ About (99-99.5) % of the electrons energy becomes heat energy and Only
- ❑ (0.5-1)% become useful X-rays



1.3 X-ray image system

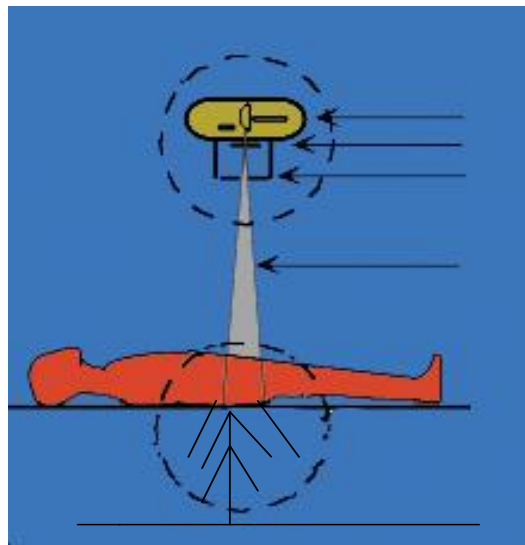
X-Ray image system

1-Production and shaping of the x-ray beam

2-Interaction and image information

3- Image capture

4-Image Visualization



X-ray tub
Filter
Collimator

X-ray field

Patient

Grid

Image Receptor

Processor



X-ray
image

1.4- FLUOROSCOPY

Components:

- Scattered radiation
- X-Ray Tube (XRT)
- X-Ray Image Intensifier (XRII)
- Video Camera

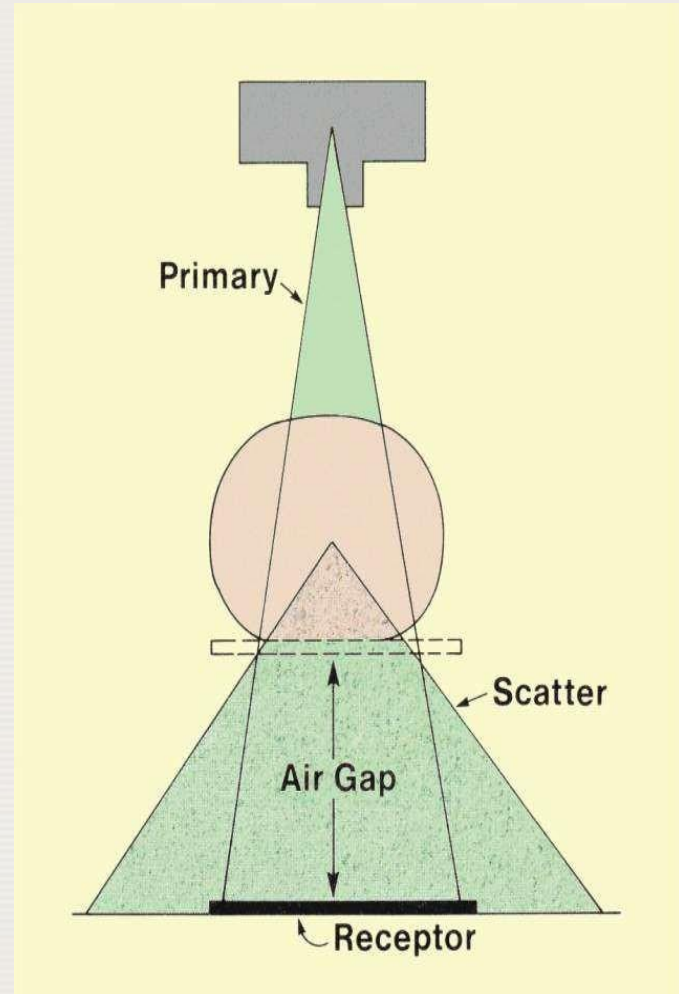
To study the basics of fluoroscopy



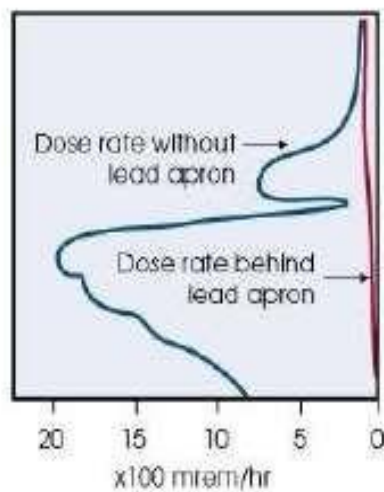
1.4 Scattered radiation

There are three types of X –rays:

- **primary X –rays** : are released from the X -ray tube
- **scattered X –rays** : are scattered from the primary substance and occur as a result of collision of electrons;
- **residual radiation** : is X -rays that pass through the patient and strengthen the image.



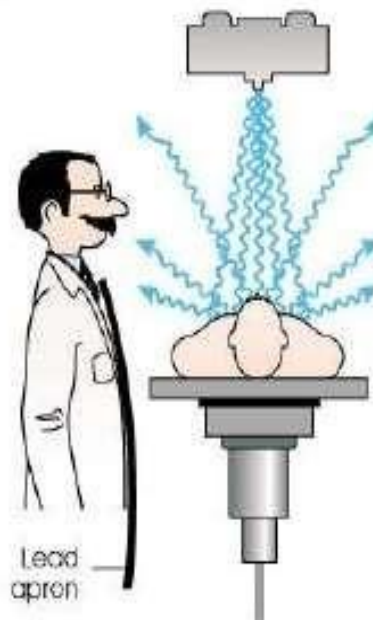
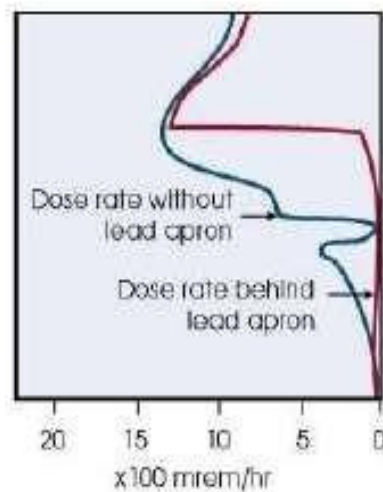
A



FLUOROSCOPY –

ABOVE AND BELOW THE TABLE TUBE EXPOSURE TO TECH

B



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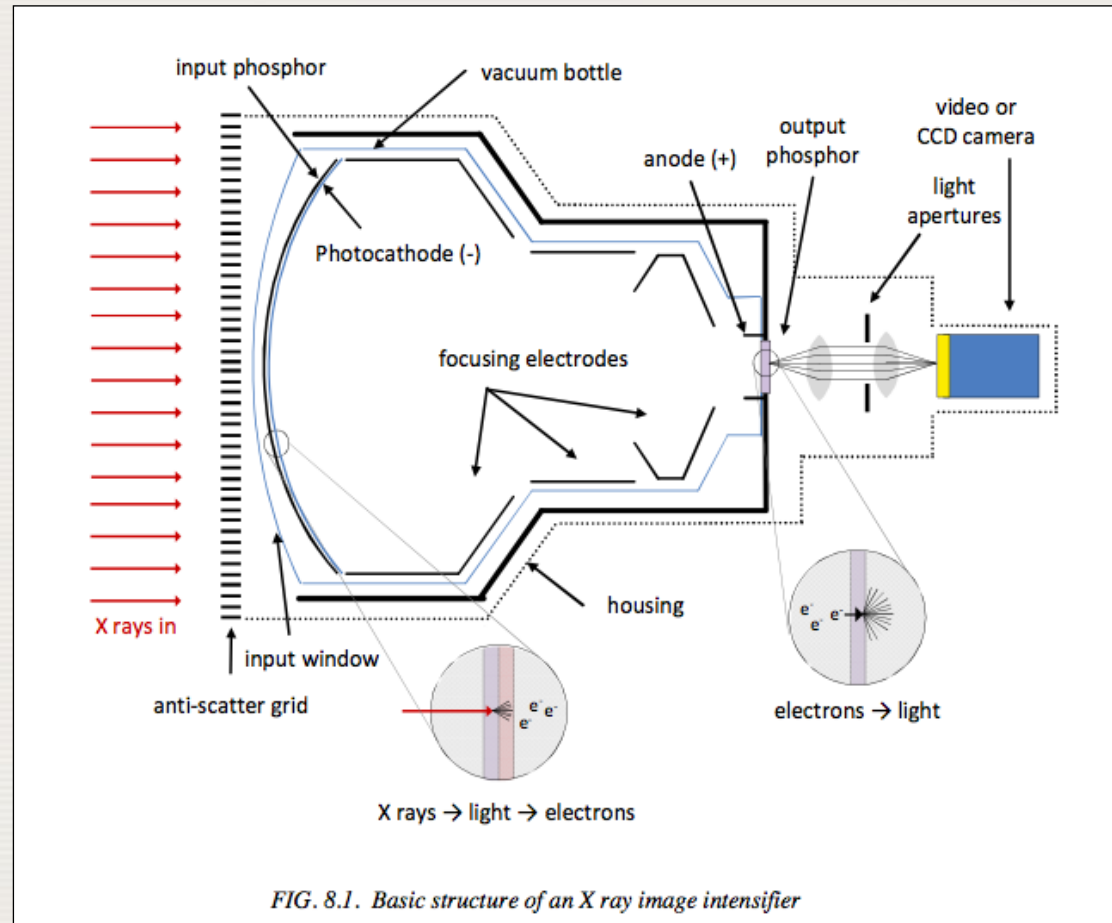


IAEA

FLUOROSCOPIC EQUIPMENT

1.4 The Fluoroscopic Imaging Chain

- **Input Phosphor**
converts: X-Rays
to Light
- **Photocathode**
converts: Light
to Electrons
- **Output Phosphor**
converts: Electrons to
Light



FLUOROSCOPIC EQUIPMENT

Noise in fluoroscopic images is high, since low IAKR (Incident X-ray Air Kerma Rate) is used to maintain patient dose at an acceptable level.

Radiography: good SNR, poor **Temporal Resolution**

Fluoroscopy: poor SNR, good Temporal Resolution

(signal-to-noise ratio) = SNR

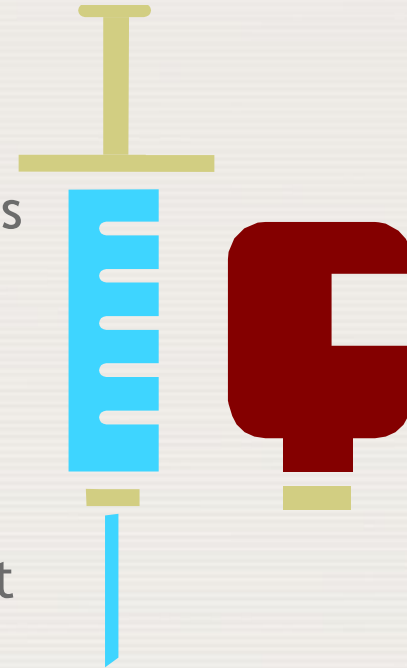
FLUOROSCOPIC EQUIPMENT

- Show continuous x-ray image
- Plays out like a movie
 - Images taken quickly allow for this to happen
- Shows movement of body parts
 - Also shows instruments or dye



FLUOROSCOPY USES

- Used in a variety of procedures
 - Examples include:
 - Orthopedic Surgery
 - Observe fractures and healing bones
 - Catheter Insertion
 - Direct catheter placement (Angiography/Angioplasty)
 - Barium X-Rays
 - Observe movement through GI tract
 - Blood Flow Studies
 - View blood flow to organs



1.5 - Mammography Machine

X-Ray Tube

Compression Paddle

Film Holder

Foot Peddles



1.5.1 Tubes, filters and spectra

- ❑ X ray spectrum should provide a range of energies that give an appropriate compromise between **radiation dose** and **image quality** for the tissues under examination
- ❑ X ray spectrum determined by **target material, filter material, and tube voltage (kV)**
- ❑ For screen-film mammography optimum beam energy lies between **18 and 23** keV depending on breast thickness and composition
 - Characteristic X rays from **molybdenum and rhodium** are suitable
- ❑ Higher energies may be more optimal for digital mammography

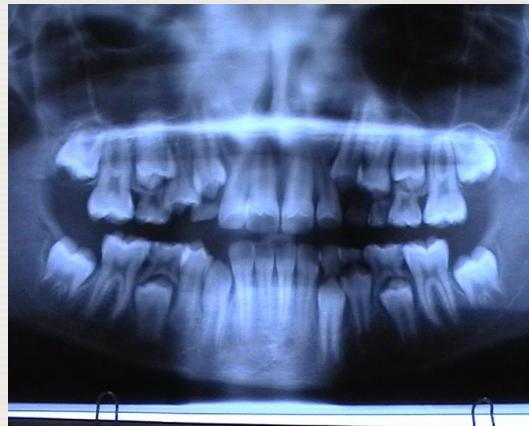
1.5.2. Compression

- ❑ Compression should be firm but not painful
- ❑ Reasons for applying compression
 - Reduces superposition of tissues
 - Decreases ratio of scattered to transmitted radiation reaching the image receptor
 - Decreases the distance from any plane within the breast to the image receptor reducing geometric unsharpness
 - Compressed breast provides lower overall attenuation allowing radiation dose to be reduced
 - Compressed breast provides more uniform attenuation over the image reducing the exposure range which must be recorded
 - Provides a clamping action reducing anatomical motion during the exposure reducing image unsharpness

1.6. DENTAL RADIOGRAPHY

□ The tooth can be imaged in three ways

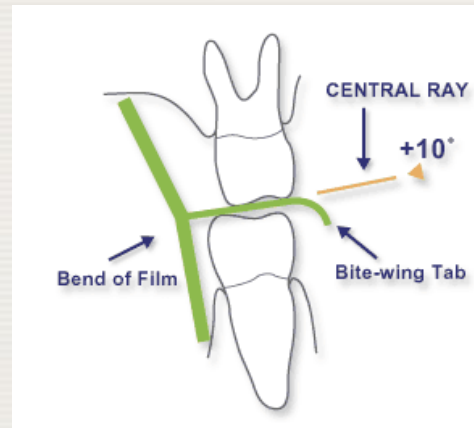
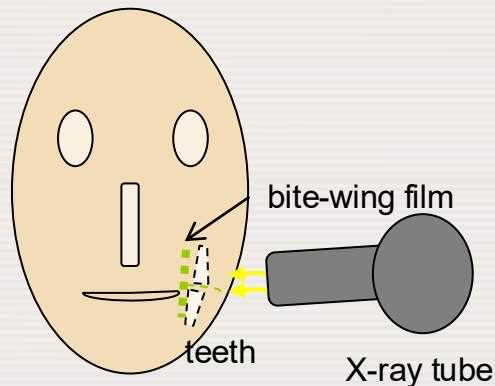
- Intra oral examination with the x-ray tube external and a bitewing film placed inside the mouth
- Extra oral examination where both the X-ray tube and detector is external to the patient to form an OPG
- A conebeam CT image



DENTAL RADIOGRAPHY

□ An intra oral examination with bite-wing

- Is the most common examination, and is a low cost technique
- Places very small demands on X ray generation since the tooth is a low attenuation static object
- The image receptor is placed inside the mouth, and irradiated externally.



DENTAL RADIOGRAPHY

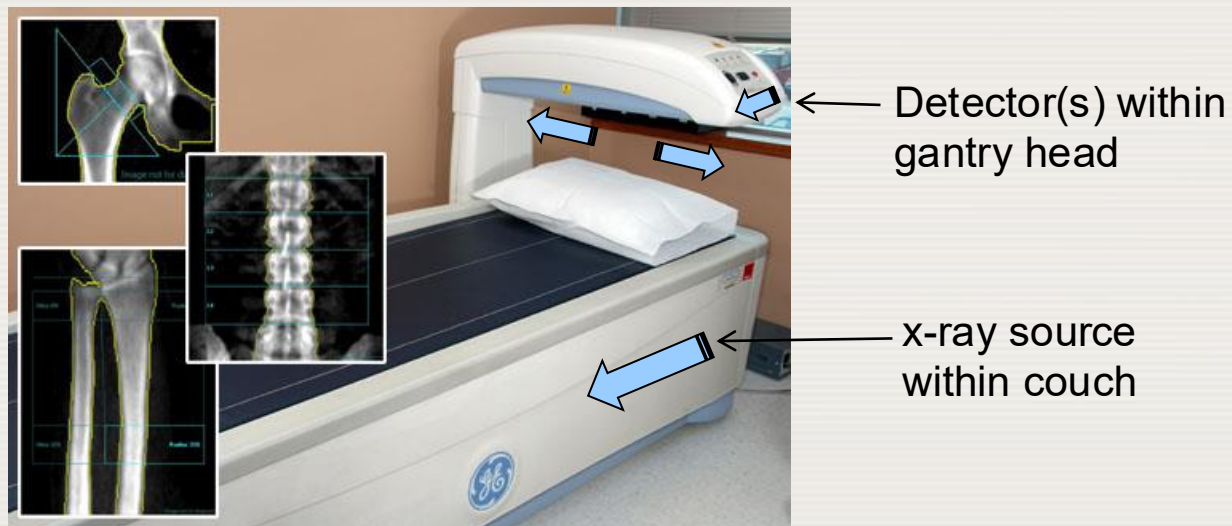
❑ Orthopantomograph (OPG),

- Two dimensional images when radiographs of the entire set of teeth are required
- The image receptor and the X ray source are external to the patient
- The X ray beam is transmitted through the head - demanding significant X ray generation power and complex motion control for the X ray tube and image receptor
- Image receptors are film or digital detectors



1.7 DUAL-ENERGY X-RAY ABSORPTIOMETRY

- ❑ The principle of operation involves **two images**
 - from the attenuation of a **low** and a **high X-ray** energy beam
- ❑ Using special imaging equipment
 - comprising of special **beam filtering** and near-perfect spatial registration of the **two attenuation maps**



DUAL-ENERGY X-RAY ABSORPTIOMETRY

- ❑ DXA can only solve for two materials simultaneously.
- ❑ However, three materials may be quantified: **bone, lean, and fat mass**, by using three fundamental assumptions,
 1. X-ray transmission through the body for the **two energy windows** can be accurately described by exponential attenuation processes
 2. Pixels of the human body image can describe two-components
 - i.e. either **soft tissue** and **bone mineral**, or, when bone is not present, fat and lean mass.
 3. **The soft tissue** overlaying the bone in the image has a composition and X-ray properties

1.8. Computed Tomography (CT)



Computed Tomography :Basic
Principle and its generations

1.8. Computed Tomography (CT)

- ❑ Clinical Computed Tomography (CT) was introduced in **1971 - limited** to axial imaging of the brain in neuroradiology
- ❑ It developed into a versatile **3D** whole body imaging modality for a wide range of applications in for example
 - oncology, vascular radiology, cardiology, traumatology and interventional radiology.
- ❑ Computed tomography can be **used** for
 - diagnosis and follow-up studies of patients
 - planning of radiotherapy treatment
 - screening of healthy subpopulations with specific risk factors.

Patient Positioning and Data Acquisition

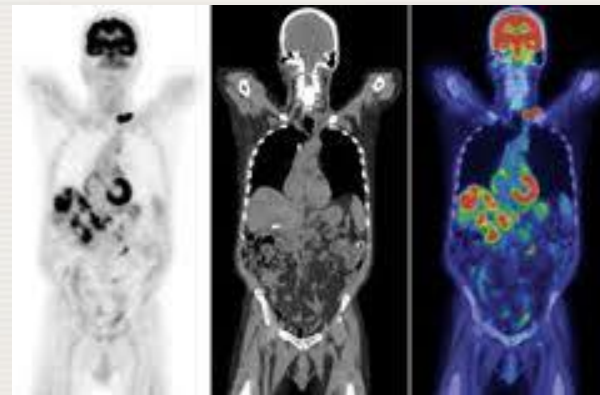


PORTABLE CT SCAN

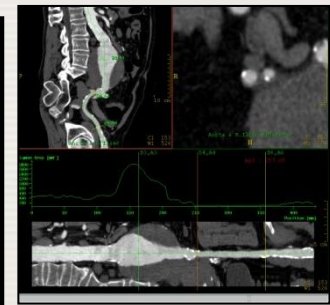
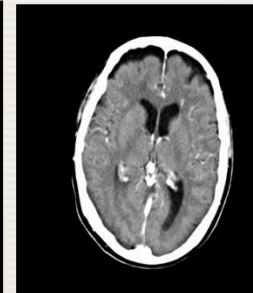


❑ Nowadays dedicated CT scanners are available for special clinical applications, such as

- For **radiotherapy planning** - these CT scanners offer an extra wide bore, allowing the CT scans to be made with a large field of view.
- The integration of CT scanners in **multi modality imaging** applications, for example by integration of a **CT scanner with a PET scanner** or a **SPECT scanner**.

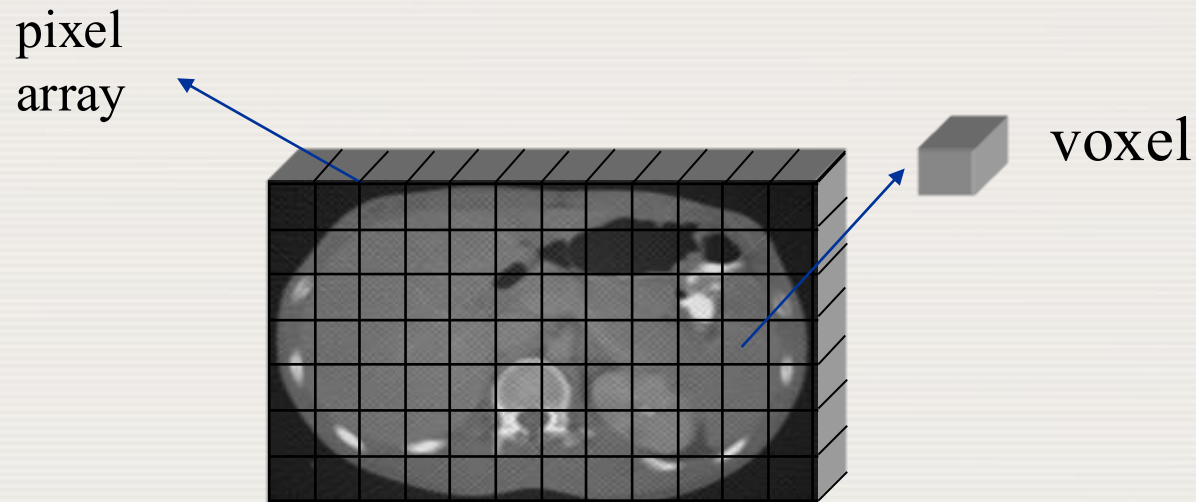


- ❑ CT scanning is perfectly suited for 3D imaging and used in, for example, **brain, cardiac, musculoskeletal, and whole body CT imaging**
- ❑ The images can be presented as impressive **colored 3D** rendered images, but radiologists usually rely more on **black and white, 2D** images, being either the 2D axial images, or 2D reformats.



. X-ray projection, attenuation and acquisition of transmission profiles

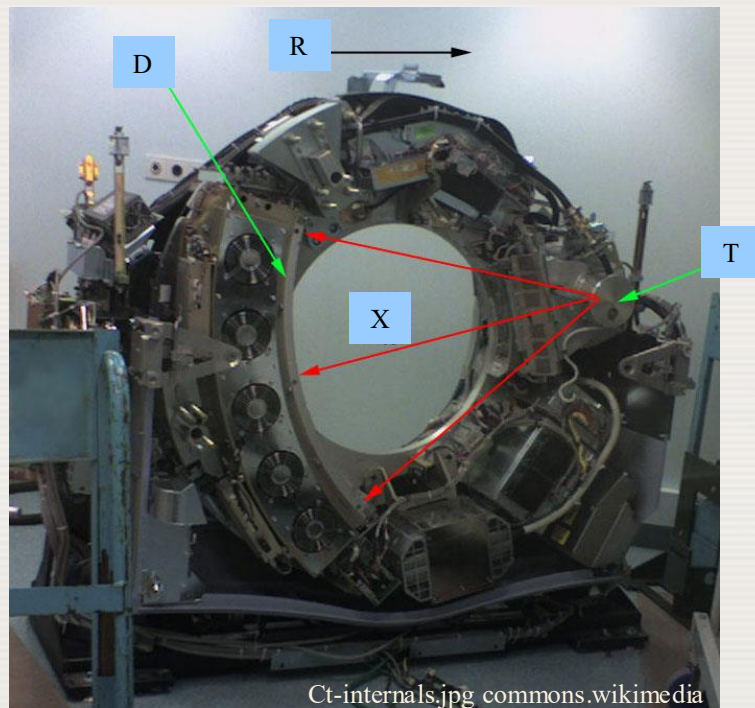
- A CT image is composed of a matrix of pixels representing the average linear attenuation co-efficient in the associated volume elements (voxels).



Gantry and table

- On the rotating part of the gantry are mounted for example
 - the X-ray tube, the detector, the high voltage generator for the X-ray tube, the (water or air) cooling of the X-ray tube, the data acquisition system, the collimator, and the beam shaping filters.

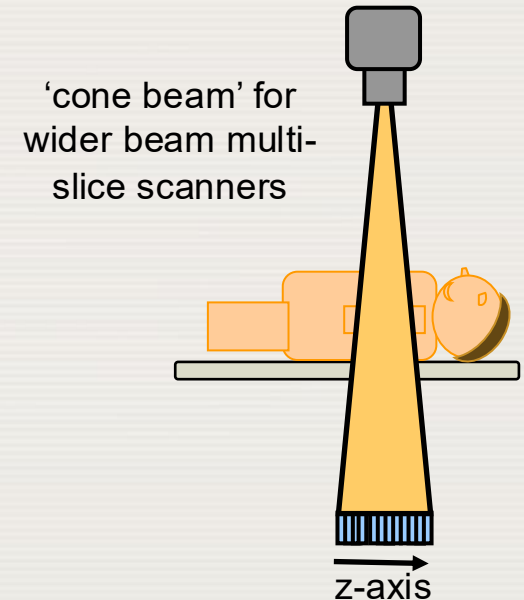
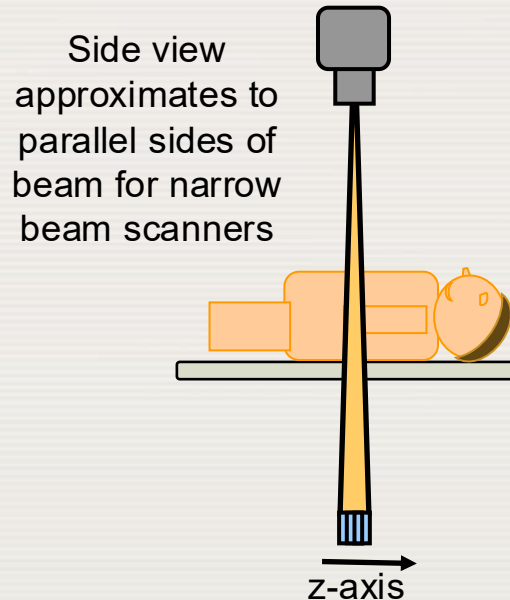
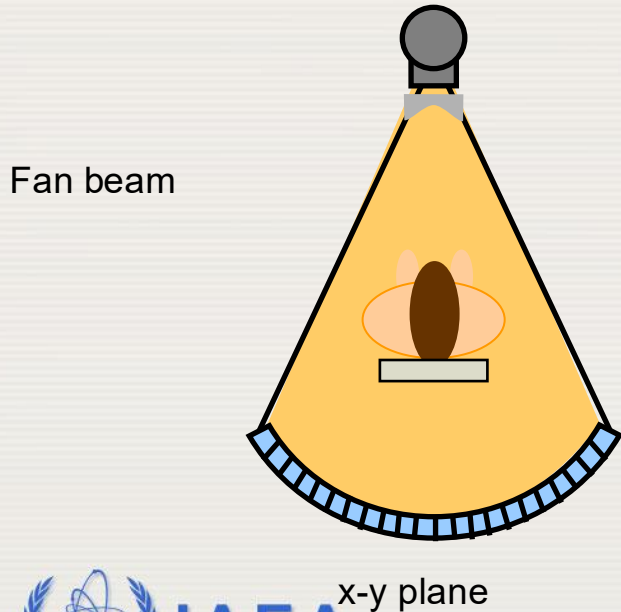
T X-ray tube
D X-ray detectors
X X-ray beam
R Gantry rotation



Ct-internals.jpg commons.wikimedia

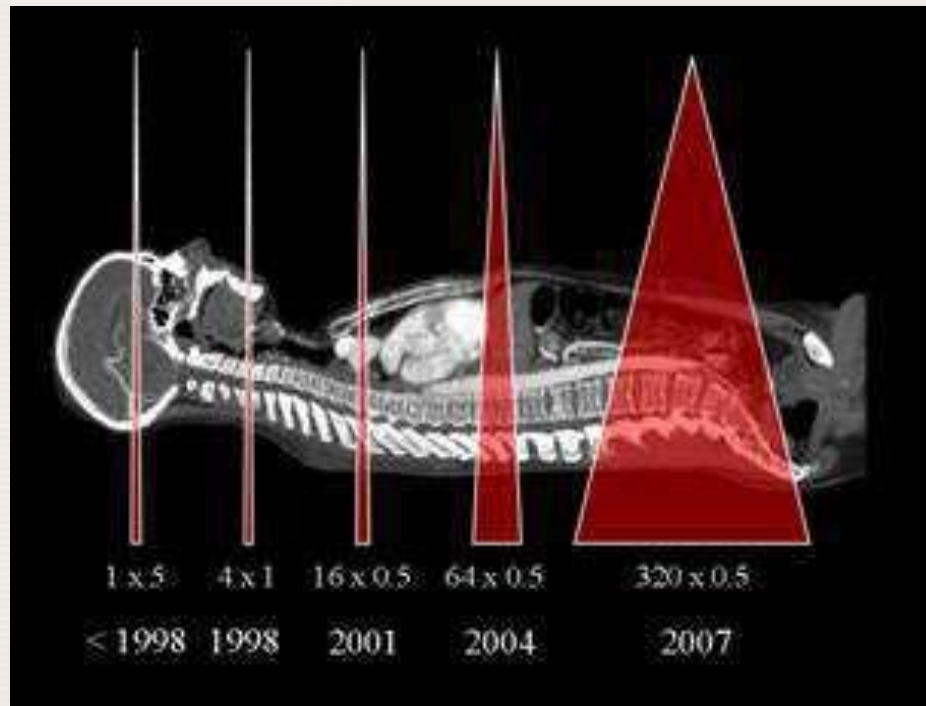
Collimation and filtration

- ❑ The X-ray beam is often referred to as a fan beam where the beam width along the longitudinal axis is small
- ❑ For multi-slice scanners where the longitudinal beam width is no longer small the X-ray beam is often referred to as 'cone beam'

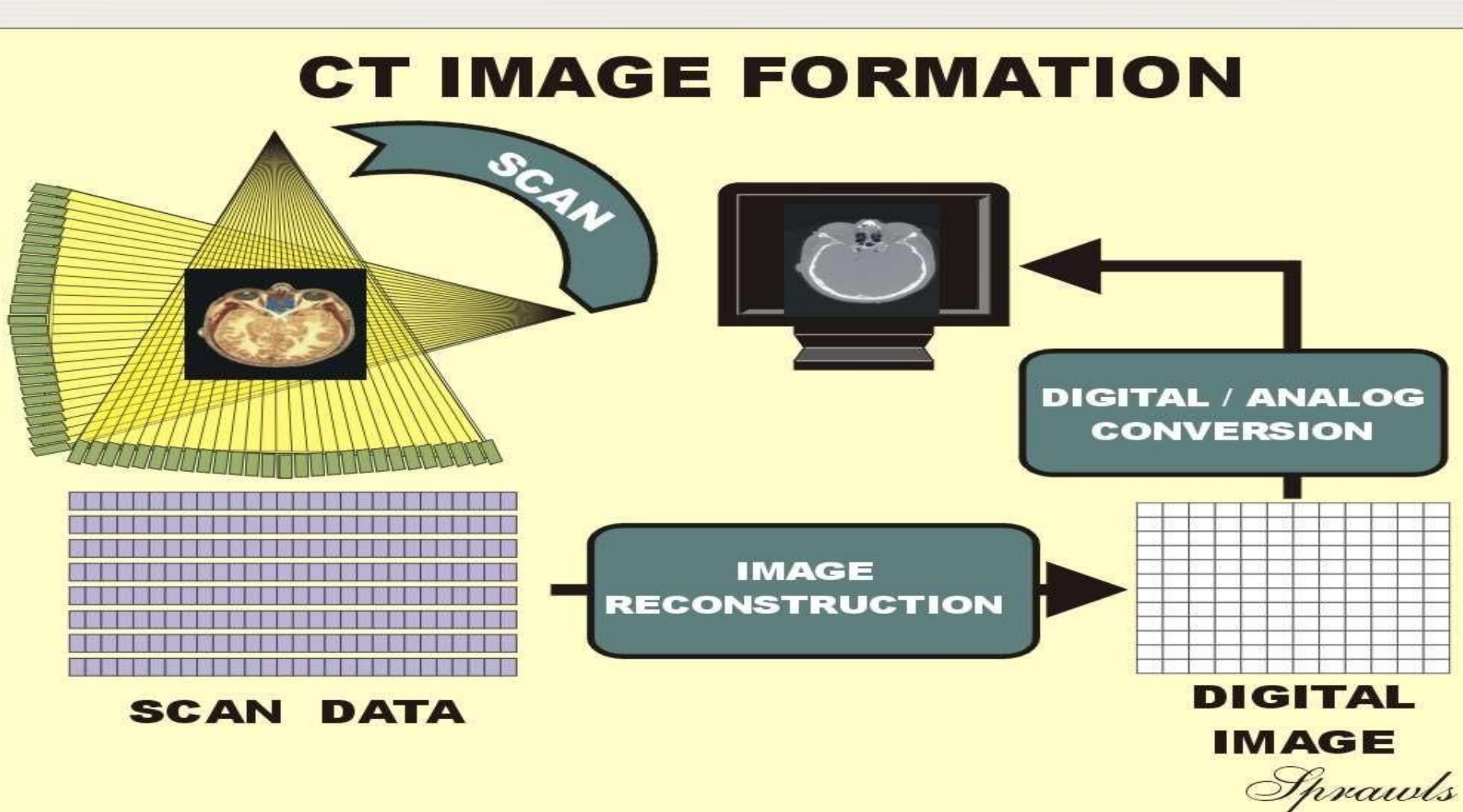


Detectors

- ❑ Increased coverage of the multi detector row CT scanners increased with more active detector rows



STEPS OF CT IMAGE FORMATION



CT Scan Methodology

- X-ray tube and detectors rotate around the patient, with the axis of rotation running from the patient's head to toe.
- 3D measurement are taken around the patient.
- Contrast dye is sometimes used to make the internal organs more visible in the image.
- Bone appears white and liquids are black; tissues are gray

Which CT Scanner is best?

☐ Axial v. Helical scanners

☐ Axial scanners

- Longer time to scan
- Danger in misregistration of scanner

☐ Helical scanners

- Quicker scan time
- Images for overlapping slices can be generated
- More complicated image reconstruction

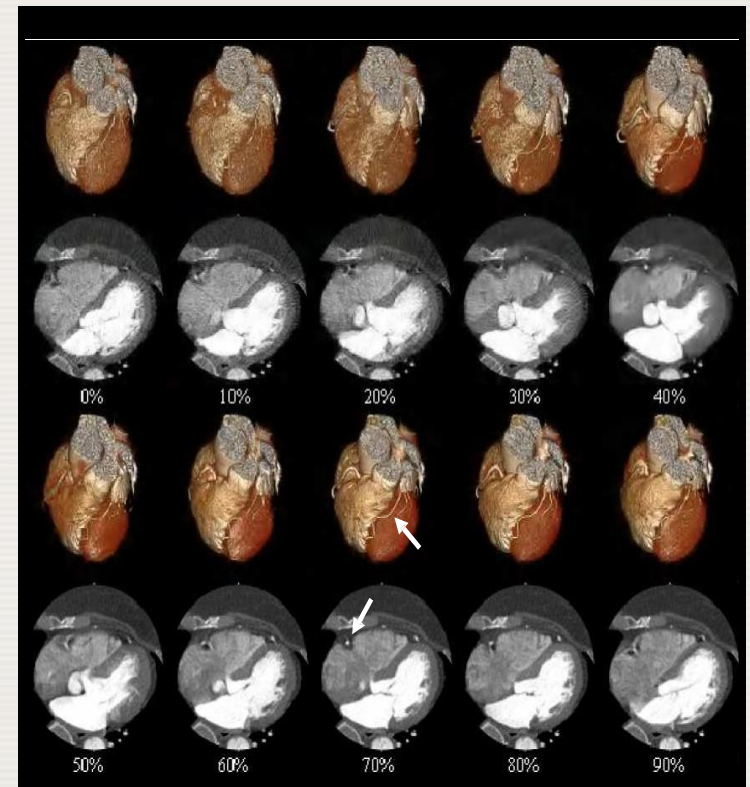
Risks of CT Scan

- ☐ Low-risk procedure
- ☐ Main risk is allergic reaction to contrast dye, often mild and resulting in itching, hives, or a rash
- ☐ More radiation than modern standard X-Ray, but still minimal amount
- ☐ Only radiation risk significant only for pregnant women (federal exposure guidelines being near zero)
- ☐ Cost is usually around \$700-\$900

Cardiac CT

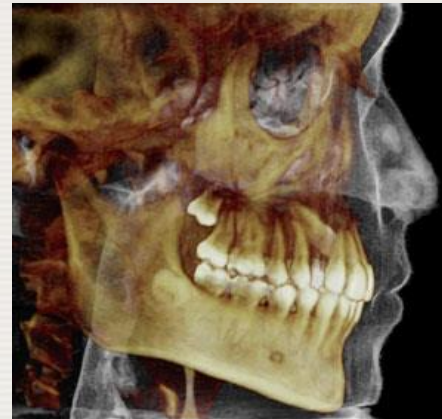
❑ Cardiac CT is based on the synchronization of image reconstruction with the ECG and selection of the best cardiac rest phase.

- The figure shows reconstructions of the heart at different cardiac phases demonstrating the difference in blurring of the coronary arteries at different cardiac phases.
- Here the cardiac phase corresponding with 70% of the RR interval produces the best, motion free, result (70% marks the start of the cardiac phase interval)



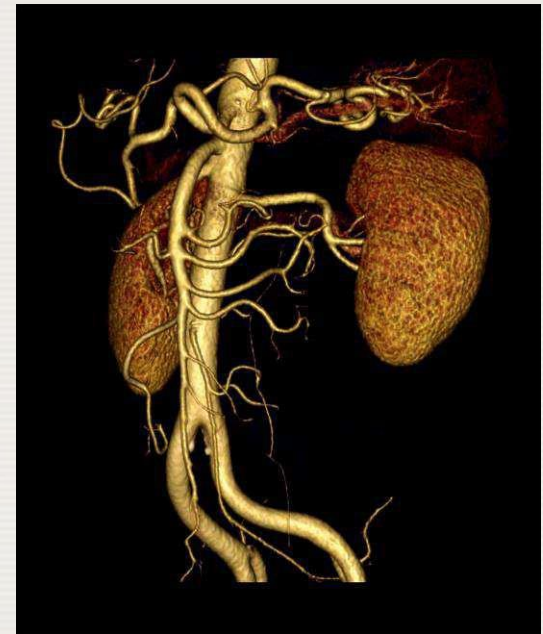
Dental CT

- ❑ CT scans of the jaw can be made with any regular CT scanner, but dedicated volume (cone beam) CT scanners for dental imaging are also available
- ❑ These are designed for the patient to be seated.



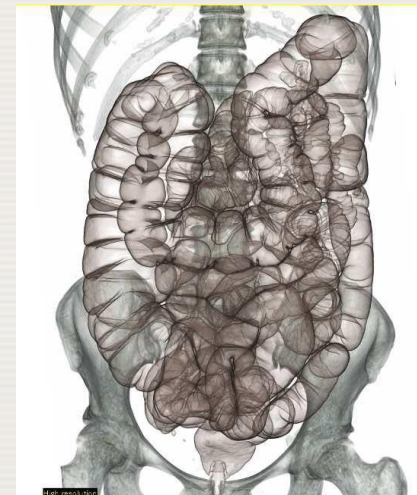
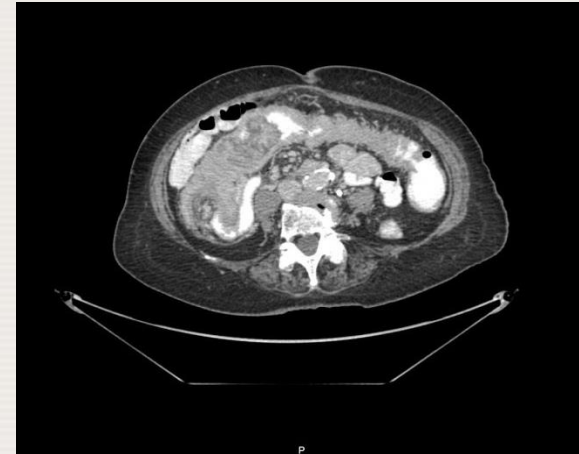
CT angiography

- ❑ In contrast enhanced CT, contrast is artificially created between structures that would not be visible on non enhanced scans.
- ❑ In CT angiography, iodine is administered during the CT scan intravenously to enhance the contrast between the vessels and the vessel wall



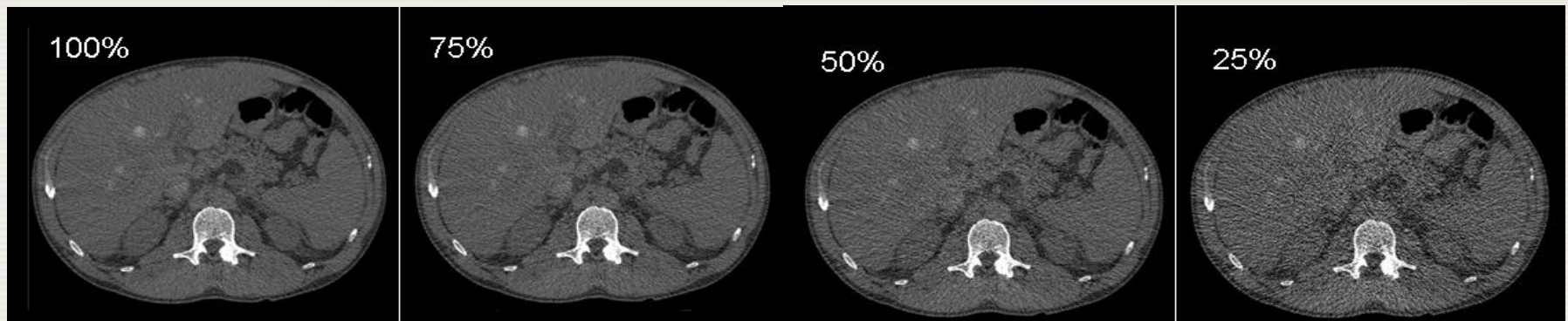
Contrast enhanced CT

- ❑ In certain studies of the abdomen, a diluted iodine solution is administered orally prior to the CT to enhance contrast within the gastrointestinal tract.
- ❑ In CT colonography gas may be inflated through the rectum to enhance contrast between the colon and its surrounding tissues



□ Example - A contrast enhanced CT scan of the liver.

- The 100% image corresponds with the actual clinical acquisition.
- Simulated noise has been added to the raw data to simulate image quality for acquisitions that are performed at 75%, 50% and 25% of the clinically used tube current.
- The appearance of the low contrast lesions in the liver becomes worse at lower tube currents due to increased noise in the images.



Calibration

- ❑ Calibrations with phantoms allow for (some) correction of the beam hardening effect

Example:

A calibration with a water filled phantom.

The relatively small phantom is used for calibration of acquisitions with a relatively small field of view.





Physics of Diagnostic Radiology -Part II

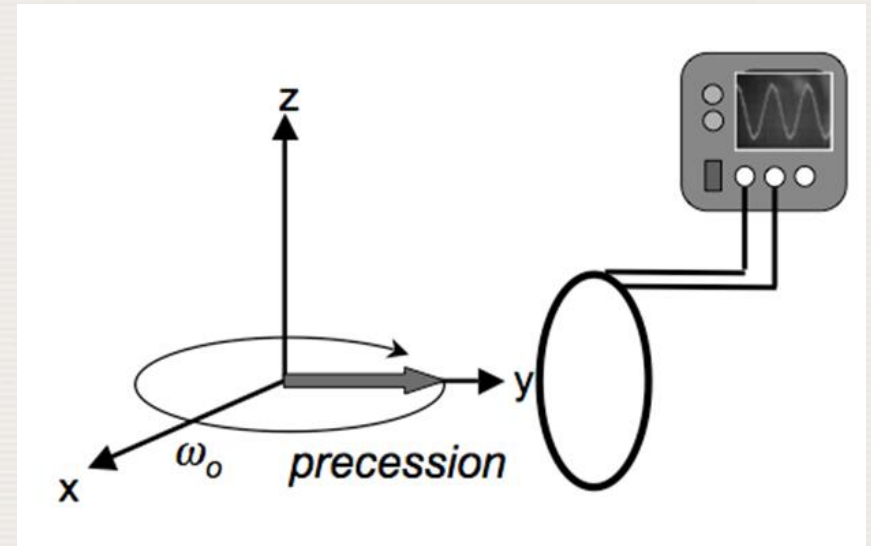
Physics of Magnetic Resonance Imaging MRI



1.9 Physics of Magnetic Resonance Imaging

Detection of a signal from rotating magnetization

- ❑ Use an RF coil
- ❑ By Faraday's law, changing magnetic flux through coil induces voltage changes
- ❑ Changes are detected by receiver



1.9 HARDWARE

Magnet designs and field strengths

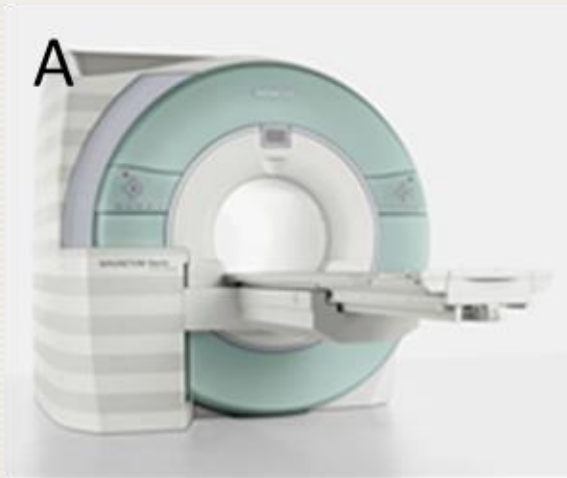
- ☐ Resistive solenoid, $< \sim 0.5$ T
- ☐ Superconducting magnet
 - clinical use: 0.5 – 3.0 T
 - experimental clinical use: 3.0 – 8 T and above
- ☐ Electromagnets, 0.1 – 0.2 T
- ☐ Permanent magnets

- ☐ Geometry relates directly to the intrinsic orientation of the magnetic field

4.9 HARDWARE

Three different superconducting magnets

- ☐ A: Siemens 3T Vario system
- ☐ B: Philips 1T Panorama system
- ☐ C: General Electric 1.5T Discovery MR450 system.



1.9 HARDWARE

The radiofrequency (RF) subsystem

□ Top row:

- Siemens 32 channel head coil
- General Electric torso array coil
- Philips neck coil



□ Bottom row:

- General Electric cervical, thoracic and lumbar spine coil



1.9 BASIC IMAGE QUALITY ISSUES

Phantoms, equipment assessment and coil loading

Phantoms for equipment assessment

□ Use for

- spatial resolution
- T_1 and T_2 relaxation time measurements
- spatial distortion assessment
- SNR measurements

1.9 BASIC IMAGE QUALITY ISSUES

Signal-to-noise ratio (SNR) and contrast-to-noise ratio

Signal-to-noise ratio (SNR)

- ❑ Signal is highly dependent on
 - object
 - imaging sequence
 - parameters chosen
- ❑ SNR must be defined for specific
 - object
 - coil
 - geometry
 - measurement sequence

4.9 MR IMAGE ACQUISITION AND RECONSTRUCTION

Dynamic Contrast Enhanced MRI (DCE-MRI)

Dynamic Contrast Enhanced MRI (DCE-MRI)

- ☐ Dynamic behaviour of a **contrast agent**
 - usually a chelate labelled with gadolinium
- ☐ Provides information about tissue function that may inform on pathology, **especially in cancer**
- ☐ Non-dynamic methods look at uptake of contrast and appearance of **lesion**
- ☐ Extension to dynamic methods
 - uptake and washout of the contrast agent

1.9 ARTEFACTS

Metal objects

Metal objects

- ☐ Ferromagnetic material can cause major distortions on the local magnetic field
- ☐ Results in
 - displacement of signals in position
 - loss of signal
- ☐ Minimise by using spin-echo sequences with short TE
- ☐ Small effects from other metallic materials
 - susceptibility artefacts leading to local distortion and loss of signal
 - conduction of currents induced by switched gradients

1.9 SAFETY AND BIOEFFECTS

- ❑ Safety issues in MR can be divided into **acute** hazards (with immediate risk of injury) and **biological** effects (with longer term effects)
- ❑ Safety concerns arise from four sources:
 - **Static field considerations**
 - projectiles, **effects on implants**, physiological effects
 - (At higher fields (> 2 T) static field can cause a degree of dizziness.)
 - **RF field considerations**
 - **tissue heating**, specific absorption rate, **burn injuries**
 - **Cardiac pacemakers** (should not be allowed in field > 0.5 mT), **hearing aids** and other magnetically controlled or programmable implants
 - **Gradient field considerations**
 - peripheral nerve stimulation, **sound pressure levels**
 - **Common MR contrast agents**

Q. / Extended Matching Questions:

For each clinical presentation below, choose the **SINGLE** most likely cause for the clinical findings from the above list of options, each option may be used once, more than once, or not at all.

A: UV (Ultraviolet).

B: US (Ultrasound)

C: Mammography

D: Laser

E : X-ray

F : CT- Scan

G: Visible Light

H: Blood pressure

I: Hearing loss

J: OCT (Ophthalmic CT-Scan)

1. A woman is a 32 year old lifelong non-smoker with no significant medical history. She presented to the emergency department with a short history of unilateral chest pain without difficulty in breathing.
2. A 74 old man admitted with a history of poor mobility and an increasing number of falls over the last week. He has a low back pain for last six week and complains of bilateral leg weakness.
3. A 33 –year old woman presented to the accident and emergency department with a headache, when she was recovering well you take his opportunity to undertake a full history and examination. On abdominal examination you found two mass.
4. A 78 year old man is a retired baggage handler and used to work at International Airport. His family has been complaining that he watches TV with the volume too loud.
5. A woman has a pain in the left side of chest after examination you found a small mass.

Thanks for lessening